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DEALER BEST PRACTICE SERIES

Use of Thermal Imaging to Inspect / Control Weld Quality

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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1.0 Introduction

Thermal Imaging Cameras have been used many years on some industries to inspect welds and control quality. Dealer Ferreyros is now using the technology in our welding operations.

It is known that temperature control is very important in welding process in order to obtain a quality product (welded joint). Because of this, Ferreyros use a thermography camera which helps us to control the temperature of preheating, post-heating and between passes in a versatile manner.

Furthermore, the use of the camera can detect defects in the weld joint; for example, surface porosity, slag inclusions and lack of penetration.

2.0 Best Practice Description

Previously, the temperature control in the welding process was performed using a pyrometer. Currently, the temperature control and the quality of the welding are performed using a thermal imaging camera.

The use of the camera helps us monitor temperatures in different processes, such as: Preheat, postheat and temperature between passes in a simple way. Also, by using the thermal imaging camera, we can detect defects present in the welded joints due to the difference of thermal gradients.

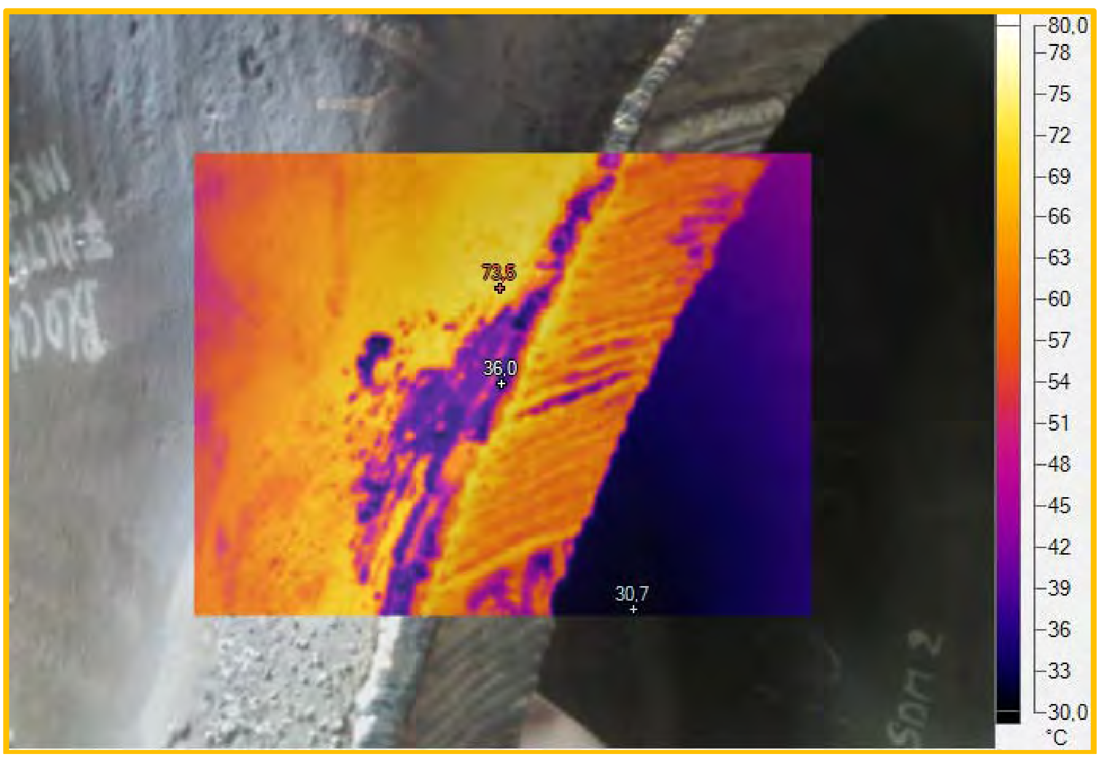

Figure 1

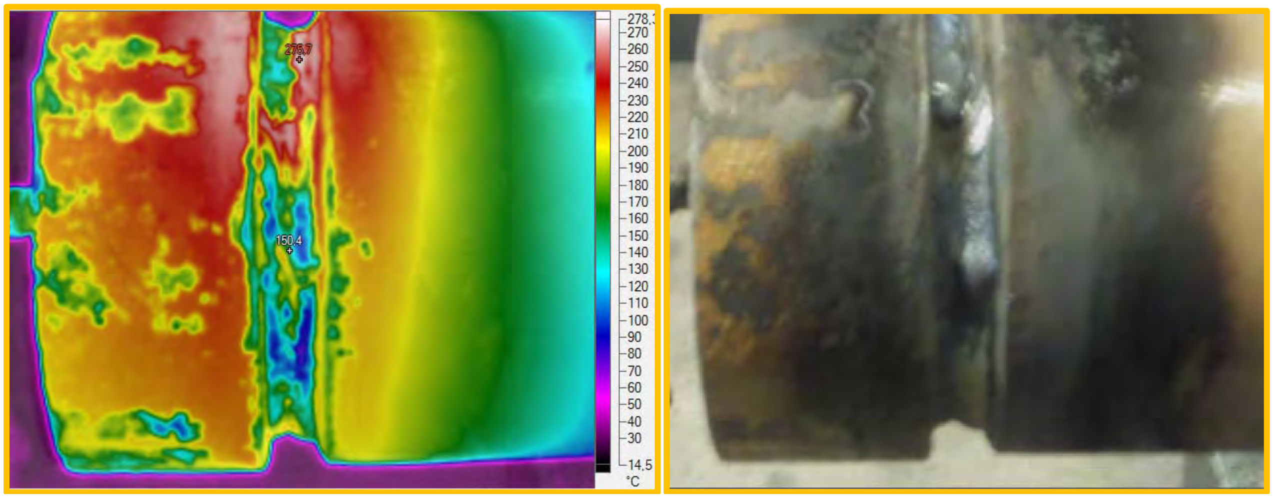

Figure 2

In the below picture we can see the temperature control in a heating process and a defect found in a welded joint.

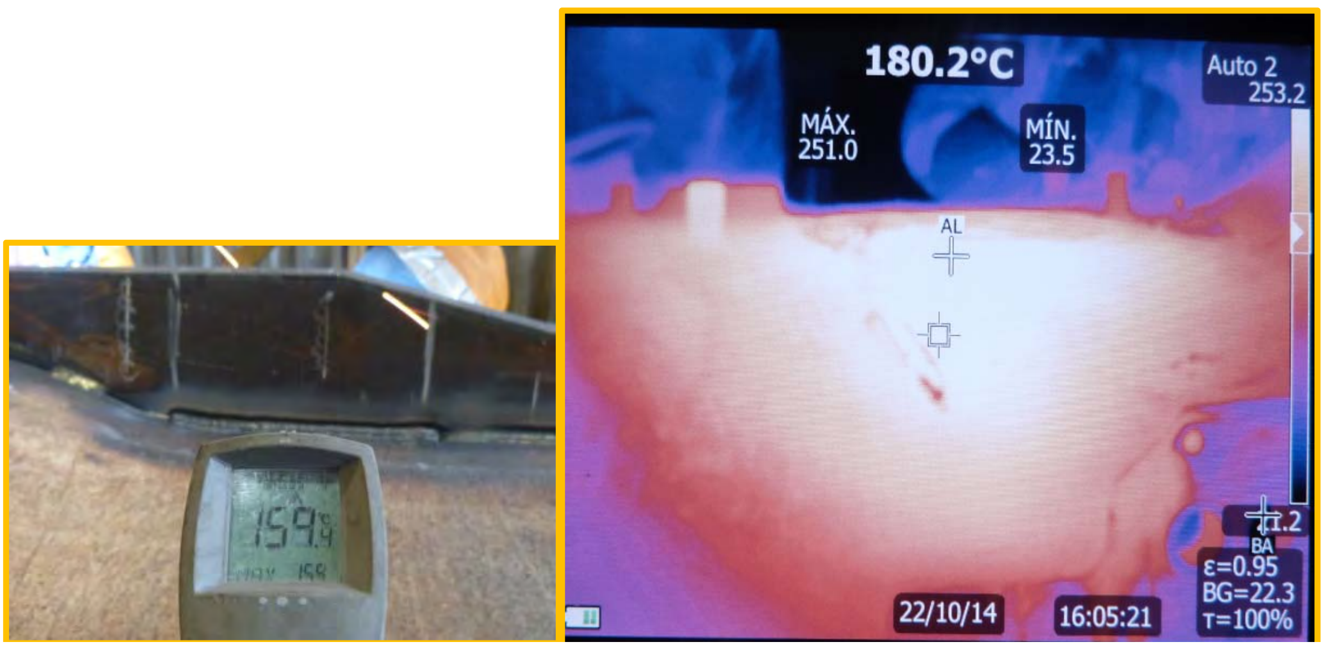

Figure 3

3.0 Implementation Steps

Before starting the process of measuring temperature, should note the following considerations:

- Type of material to be inspected.
- Lighting conditions, according to the inspection site.
- Technical information or location maps of the areas measured.
- Areas of measurement properly identified.

i. Temperature control

a) Set the following parameters:

- Emissivity depends on the material to be heated.
- Temperature range depends on the temperature at which the component is going to be preheated, post-heating.

b) Approximate the thermography camera and proceed to take data for temperature control.

ii. Determination of defects in welds

The technique is applied during the cooling of the part, immediately after application of the weld and this depends on the temperature gradients present in the weld bead.

In the thermogram shown in figure 4, we can see a yellow elongated indication on the right side of the end, indicating that these points or areas are colder than the weld cord.

This area, to be at a cooler temperature than the rest of the cord gives an indication of heat dissipation at these points, that is, there is an interruption in the thermal conductivity of the material. This area coincides with the lack of weld penetration. That empty space or loss of continuity of metal, explains why in the thermogram is seen as a cold spot.

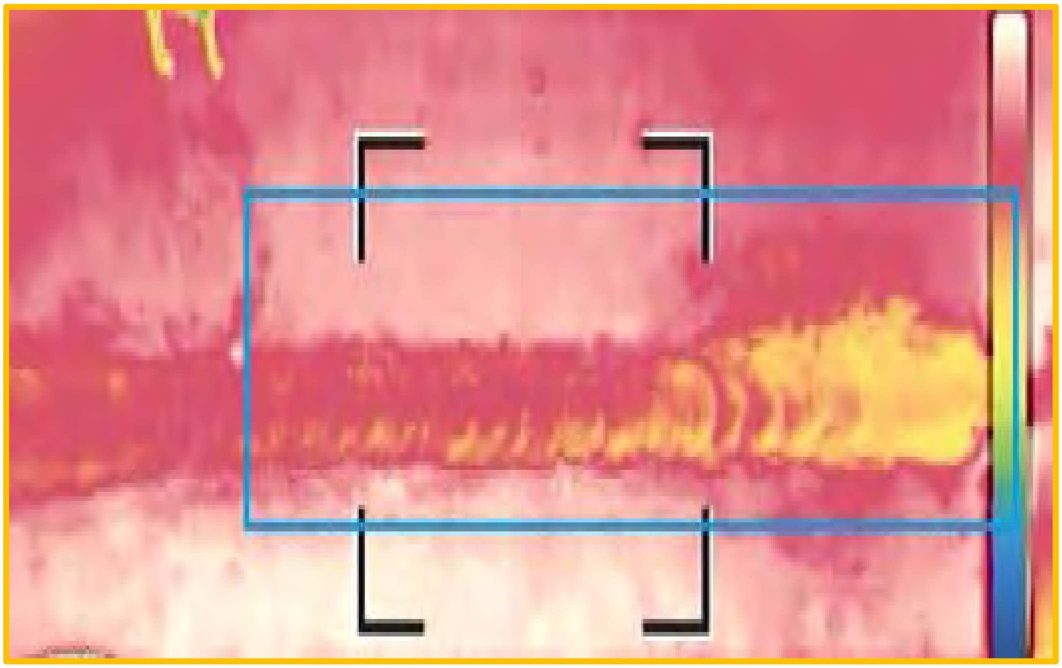

Figure 4

4.0 Benefits

- Have a better temperature control in the welding process (dynamic control).
- Find welding defects – prevent failures
- Quick and easy method of inspection.

5.0 Resources Required

- Thermal Imaging camera equipment.
- This technology is well documented on uses of thermal imaging cameras. Search on the internet for "Thermal Imaging Weld Inspection" and you will find guidelines

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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